

STAT 7900: SPECIAL TOPIC: CAUSAL ANALYSIS



INSTRUCTOR

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STAT 7900

Final Project

Slide: Final Project Topics

- **Observational Causal Analysis**: Estimate the causal effect of an exposure or treatment using real observational data and appropriate adjustment methods.
- **Heterogeneous Treatment Effects**: Investigate how causal effects vary across individuals or subgroups.
- **Instrumental Variables / Natural Experiments**: Estimate causal effects when treatment is confounded using a defensible instrumental-variable strategy.
- **Policy Evaluation**: Evaluate the causal impact of a policy using Difference-in-Differences, event-study, or synthetic-control methods.
- **Causal Discovery**: Use observational data to learn and evaluate plausible causal structures and relationships.

Students may work individually or in teams of two. Applied, methodological or theoretical projects are all welcome.

Final Project

Deadline	Deliverable
October 12	Project Proposal: topic, dataset, causal question/objective, treatment, outcome, target population and proposed approach.
November 9	Introduction & Literature Review: motivation, existing literature, research gap and updated analysis plan.
December 7	Final Project & Presentation: complete paper, reproducible analysis/code and presentation.

Goal: Develop the project in the format of a research study; particularly strong projects may be suitable for further development toward publication.

Questions

- A model predicts heart attack with $AUC = 0.94$ using smoking status. Does that establish a causal effect?
- Why can Simpson's paradox occur?
- Write one predictive and one causal question about college GPA.

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Because a third variable can affect both group membership and the outcome, causing aggregate and within-group associations to differ.

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Because a third variable can affect both group membership and the outcome, causing aggregate and within-group associations to differ.

- Write one predictive and one causal question about college GPA.

Predictive: Can study hours predict college GPA? **Causal:** Would increasing study hours improve college GPA?

Recap

Prediction

Can a pattern predict the outcome in new data?

High predictive accuracy may reflect confounding, selection, or consequences of treatment.

Causation

What would change if we intervened?

A causal claim requires a counterfactual comparison plus assumptions.

Recap

The scientific question determines the analysis.

Association: $E[Y \mid T = 1] - E[Y \mid T = 0]$

Causal effect: $E[Y(1)] - E[Y(0)]$

These two quantities can be very different.

What is Causal Effect?

A causal effect compares two possible worlds that differ in the treatment.

World 1

The unit receives treatment.

$$T = 1$$

World 0

The same unit does not.

$$T = 0$$

Example 1

Does tutoring help Sarah?

Sarah attends tutoring and scores 88 on the final exam.

We cannot answer from 88 alone.

We also need to know what Sarah would have scored without tutoring.

Tutoring: $Y(1) = 88$, No tutoring: $Y(0) = ?$

Individual causal effect $= Y(1) - Y(0)$

Example 2: Medicine

Drug

$Y_i(1)$ = Systolic BP if patient i
receives the drug

No Drug

$Y_i(0)$ = Systolic BP if same
patient does not

If $Y_i(1) = 128$ and $Y_i(0) = 142$, then the causal effect is -14 .

Example 3: Policy

Would raising a state minimum wage change employment next year?

$Y(1)$ = Employment under the wage increase

$Y(0)$ = Employment without the wage increase

The state cannot live through both versions of the same year.

Example 4: Sports

Would the team have won if the red card had not occurred?

Observed Match

Red card occurred.

Team Lost

Counterfactual Match

No red card

What would have happened?

Potential Outcomes

$$T_i \in \{0,1\}$$

$Y_i(1)$ = Outcome unit i would have under treatment

$Y_i(0)$ = Outcome unit i would have under control

Causal effect: $\tau_i = Y_i(1) - Y_i(0)$

We observe only the potential outcome corresponding to the treatment received.

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$$

If $T_i = 1$, $Y_i = Y_i(1)$ and if $T_i = 0$, $Y_i = Y_i(0)$.

Potential Outcomes

Binary treatment is just the simplest case.

- Multi-level treatment: $Y_i(0), Y_i(1), Y_i(2), \dots$
- Continuous Treatment: $Y_i(t)$ for dose t .
- Time-Varying treatment: Potential outcome depends on a treatment history.
- Dynamic regimes: Treatment can depend on previous covariates or outcomes.

We start with $T \in \{0,1\}$ because the causal logic is simplest there.

Individual Causal Effects

Person	Y(1)	Y(0)	τ
A	80	70	+10
B	72	72	0
C	60	75	-15
D	90	80	+10

$$\tau_i = Y_i(1) - Y_i(0)$$

Individual Causal Effects

We can have same treatment, different responses.

“Does the treatment work?” is often an incomplete scientific question.

What should causal questions specify?

- Population: For whom?
- Treatment: Which intervention?
- Comparator: Compared with what?
- Outcome: What is being measured?
- Time: Over what follow-up period?

What do we actually observe?

Person	T	Y(1)	Y(0)	Observed Y
Anna	1	72	?	72
Ben	0	?	64	64
Carlos	1	81	?	81
Diana	0	?	75	75

Every person is missing one potential outcome.

Fundamental problem of causal inference

For each unit, we can observe $Y_i(1)$ or $Y_i(0)$.

Never both at the same time.

Observed: Factual Outcome.

Missing: Counterfactual Outcome.

Fundamental problem of causal inference

A causal inference problem is a missing-data problem.

Observed: (T_i, Y_i, X_i)

Missing: $Y_i(1 - T_i)$

The missingness is structural: every unit is missing exactly one potential outcome.

We cannot recover the missing value for an individual without very strong assumptions.

But we may identify averages by comparing suitable groups.

How can Causal Inference be possible?

We stop trying to estimate individual causal effects.

Instead, we target population averages that can be identified under assumptions.

$$\tau_i \rightarrow E[\tau_i]$$

Average Treatment Effect (ATE):

$$ATE = E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)]$$

Average effect if the target population were treated versus if the same target population were untreated.

ATE answers a population-level intervention question.

ATE, ATT, ATC

Average Treatment Effect: $ATE = E[Y(1)] - E[Y(0)]$

Average Treatment Effect on treated: $ATT = E[Y(1) - Y(0) | T = 1]$

Among people who received treatment, how much did treatment change the outcome?

Average Treatment Effect on Controls: $ATC = E[Y(1) - Y(0) | T = 0]$

Among untreated patients, what would happen on average if they receive treatment?

Conditional Average Treatment Effect (CATE)

$$\text{CATE}(x) = E[Y(1) - Y(0) \mid X = x]$$

Example

$$\text{CATE}(\text{age} > 65) = -10$$

mmHg

Another subgroup

$$\text{CATE}(\text{age} < 40) = -2$$

mmHg

This is the language of treatment-effect heterogeneity.

Different estimands answer different questions

Estimand	Target population	Question
ATE	Everyone	What if everyone were treated vs untreated?
ATT	Actually treated	What was the effect among treated units?
ATC	Actually untreated	What would treatment do for untreated units?
CATE(x)	Units with $X=x$	How does the effect vary by covariates?

Worked Examples

i	T	Y(1)	Y(0)	τ
1	1	80	70	10
2	0	72	72	0
3	1	60	75	-15
4	0	90	80	10
5	1	83	78	5
6	0	65	60	5

Worked Examples

i	T	Y(1)	Y(0)	τ
1	1	80	70	10
2	0	72	72	0
3	1	60	75	-15
4	0	90	80	10
5	1	83	78	5
6	0	65	60	5

$$ATE = 2.5$$

$$ATT = 0$$

$$ATC = 5$$

Worked Examples

$$ATE = P(T = 1) \cdot ATT + P(T = 0) \cdot ATC$$

In the previous example, half the population is treated and half untreated.

$$0.5(0) + 0.5(5) = 2.5$$

Bias Decomposition

$$E[Y | T = 1] - E[Y | T = 0]$$

$$= ATT + \{E[Y(0)|T = 1] - E[Y(0)|T = 0]\}$$

So, Observed treated control difference =
ATT + Selection Bias

The treated and untreated groups can differ for two reasons: treatment changed the outcome, and they were different even before treatment.

Bias Decomposition: Example

Suppose: people who get treatment has an average outcome of 80 and untreated group has 70.

$$E[Y|T = 1] - E[Y|T = 0] = 80 - 70 = 10$$

But suppose the people who got treatment would already have had an average outcome of 76 even without treatment.

$$E[Y(0)|T = 1] = 76$$

Then: $ATT = 80 - 76 = 4$ and Baseline/Selection Difference: $76 - 70 = 6$.

So the observed 10-point difference makes the treatment look like it helped by 10, but only 4 points were actually treatment effect.

Bias Decomposition: Example

Suppose: people are divided into two groups, exercise and no exercise.

Observed mean weight loss: Exercise = 15 lbs, No exercise = 6 lbs.

Naive difference = 9 lbs

Should we conclude that the exercise program causes 9 lbs of additional weight loss?

Not yet. Check how treatment was assigned.

Bias Decomposition: Example

Age group	Exercise	No exercise
< 40	80 people; mean loss = 17 lbs	20 people; mean loss = 14 lbs
≥ 40	20 people; mean loss = 7 lbs	80 people; mean loss = 4 lbs

Within each age group: effect = 3 lbs

Standardization

What if both groups had the same age distribution?

$$\text{Treated Mean: } 0.5(17) + 0.5(7) = 12$$

$$\text{Control Mean: } 0.5(14) + 0.5(4) = 9$$

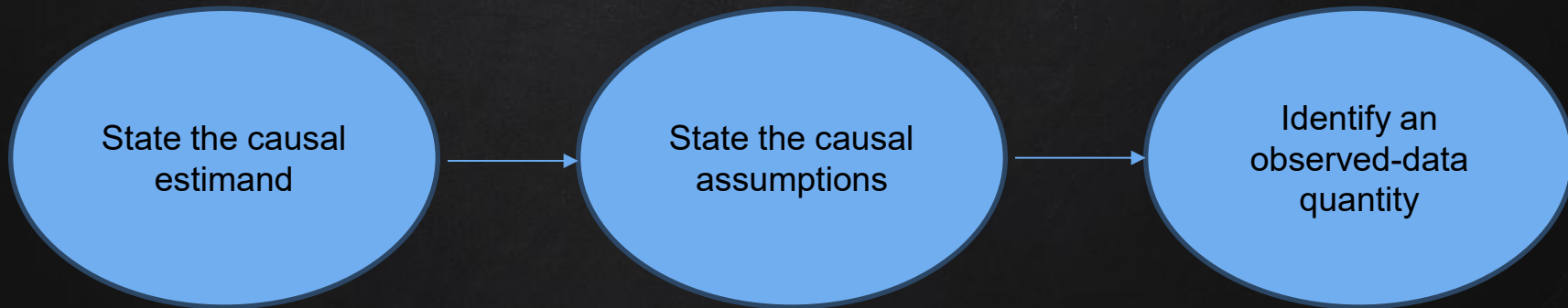
So, the standardized treatment effect is 3 lbs, matching the within-age-group effect.

Adjustment asks: what would the groups look like under a common covariate distribution?

How do we recover Causal Effect

We saw that the observed treated-untreated difference can mix a causal effect with baseline differences.

Now we ask what must be true for observed data to identify the missing counterfactual comparison.



Identifiability

Causal quantity: $E[Y(1) - Y(0)]$

Statistical quantity: $E[Y|T = 1] - E[Y|T = 0]$

Statistical quantities can be estimated from the data, causal estimands cannot.

A causal quantity is identifiable if we can compute it from a purely statistical quantity (e.g. $E[Y|t]$).

For a causal quantity to be identifiable, we need some assumptions.

Standard Causal Assumptions

For the standard potential-outcomes framework, we need:

- Consistency
- Exchangeability
- Positivity
- No interference / SUTVA

Exchangeability

Exchangeability or Ignorability means that the treated and untreated groups are comparable with respect to their potential outcomes.

If treatment assignment were swapped between the groups, we would expect the same distribution of potential outcomes.

$$(Y(1), Y(0)) \perp\!\!\!\perp T$$

Before treatment, randomized groups are comparable in expectation, including on unmeasured baseline causes.

Conditional Exchangeability

$$(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$$

Within levels of measured covariates X , treatment assignment behaves as if random with respect to the potential outcomes.

Also sometimes referred to as unconfoundedness or no unmeasured confounding.

Exchangeability is not a model diagnostic

Can we test “no unmeasured confounding” from the observed data?

No.

We can check balance on measured covariates.
But an unmeasured confounder is, by definition, not in the dataset.

Exchangeability: Example

Suppose sicker patients are more likely to receive a new drug.

Treated patients: More severe disease, worse baseline prognosis.

Untreated patients: Less severe disease, better baseline prognosis.

Even without treatment, the two groups would have had different outcomes.

Therefore: $(Y(1), Y(0)) \perp T$ assumption is violated.

Methods that can help: Regression adjustment, stratification, matching, propensity-score methods, IPW, doubly robust estimators, sensitivity analysis, instrumental variables, difference-in-differences, negative controls, etc.

Consistency

$$\text{If } T_i = t, \text{ then } Y_i = Y_i(t)$$

The observed outcome under the treatment actually received equals the corresponding potential outcome.

Consistency requires a well-defined treatment.

The treatment label must correspond to the intervention whose effect we want.

Consistency

Same Label

Exercise = 1

Many versions

Walking 20 min/week

Running 5 days/week

Resistance training

Supervised rehab

Potential outcomes require us to ask: which intervention exactly?

Consistency: Example

Example 1: Surgical intervention – A study defines exposure as $T = 1$ for receiving heart surgery. The surgical procedure groups together procedures done by world class chief surgeon with procedures done by a resident on their first shift.

Example 2: An economic model evaluates the causal effect of holding a college degree ($T = 1$). The degree category treats a STEM degree from a top-tier university exactly the same as a humanities degree from an unaccredited online college.

Example 3: A nutrition study studies the effect of a low-carb diet ($T = 1$). The category lumps together a healthy low-card diet rich in leafy greens and wild fish, with an unhealthy low-card diet relying heavily on processed meats and deep fried fast food.

SUTVA and Interference

$Y_i(t)$ depends on unit i 's treatment — not someone else's treatment

No interference: one unit's treatment does not change another unit's outcome.

No hidden versions: treatment “1” represents a sufficiently well-defined intervention.

SUTVA: Stable Unit Treatment Value Assumption

SUTVA and Interference

Setting	Why interference occurs
Vaccination	My vaccination can reduce your infection risk.
Classroom intervention	Treated students can share material with untreated classmates.
Social networks	Information given to one person can spread through friends.
Neighborhood policy	Changing one area can move traffic or crime to another area.

Remedies: Cluster randomization, partial-interference models, spillover-effect models, network causal inference, exposure mapping, etc.

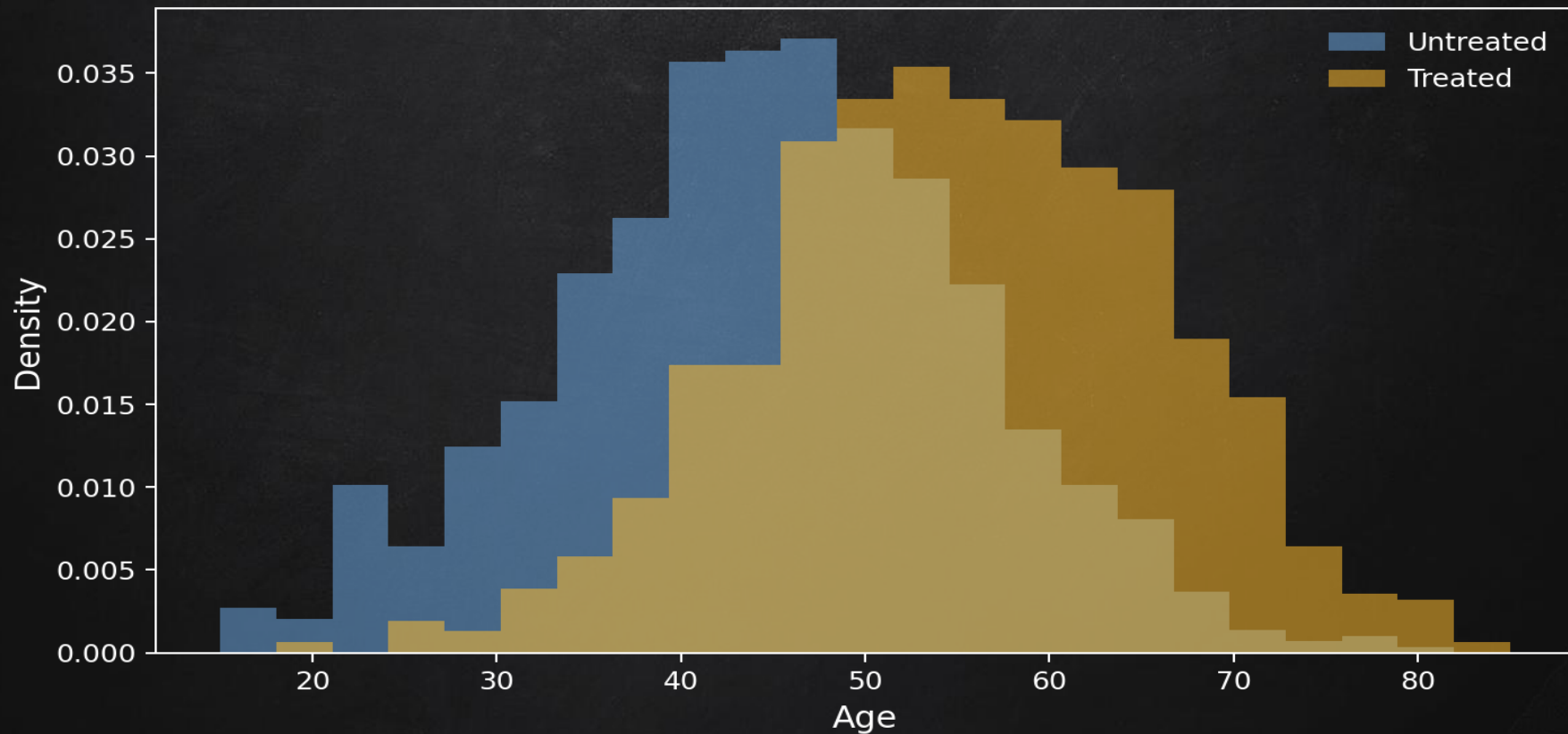
Positivity

$$0 < P(T=1 \mid X=x) < 1$$

For every covariate pattern in the target population, both treatment options must be possible.

Otherwise, we have no empirical comparison for some covariate values.
The analysis must extrapolate beyond the support of the data.

Positivity as Overlap



Positivity Violation

Suppose every patient older than 90 receives the treatment.

$$P(T=0 \mid \text{age} > 90) = 0$$

What is the untreated outcome distribution for 95-year-olds?

The observed data do not tell us. Any answer requires extrapolation or a different target population.

Positivity-Unconfoundedness Tradeoff

A richer adjustment set can help one assumption while hurting another.

MORE COVARIATE
ADJUSTMENT

Can make conditional
exchangeability
more plausible

TRADEOFF



BUT...

Can reduce overlap and
create
near-positivity violations

Goal: control the confounding that matters without
demanding comparisons the data cannot support.

Positivity-Unconfoundedness Tradeoff

Suppose treatment is surgery.

Covariates conditioned on	Example treatment probability
Age	$P(T = 1 X) = 0.55$
Age + severity	0.80
+ comorbidity	0.95
+ prior treatment + SES	0.99

Exchangeability: richer X may better account for confounding.

Positivity: richer X may leave almost no treated or untreated comparison subjects within some strata.

Positivity-Unconfoundedness Tradeoff

Adjust for AGE only

Good overlap

But disease severity and comorbidity still affect both treatment choice and outcome.

→ Residual confounding

Add SEVERITY + COMORBIDITY

Exchangeability becomes more plausible.

But among the sickest patients, nearly everyone receives treatment.

→ Weak overlap

Add still more predictors

Some covariate profiles may have no untreated patients at all.

The ATE for the original population may no longer be empirically supported.

→ Redefine target?

Positivity-Unconfoundedness Tradeoff

1

CHOOSE COVARIATES BECAUSE OF THE CAUSAL STRUCTURE

Adjustment is about blocking confounding, not maximizing the number of predictors.

2

INSPECT OVERLAP AFTER DEFINING THE ADJUSTMENT SET

Look for regions where treatment assignment is nearly deterministic.

3

AVOID UNSUPPORTED EXTRAPOLATION

If some patients have no comparable treatment alternative, model-based estimates rely on assumptions outside the data.

4

SOMETIMES CHANGE THE ESTIMAND OR TARGET POPULATION

Restrict to common support, trim extreme regions, or target an overlap population when scientifically appropriate.

Assumption Check

- A drug is never prescribed during pregnancy.
- People who feel healthier are more likely to exercise, but baseline health was not measured.
- “Diet = 1” includes keto, vegan, Mediterranean, and intermittent fasting with no further detail.
- Students assigned to a tutoring program share the worksheets with untreated classmates.

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Conditional Exchangeability.

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Consistency.

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No Interference / SUTVA.

Assumption to Identifiability

$$\begin{aligned} ATE &= E[Y(1) - Y(0)] \\ &= E[Y(1)] - E[Y(0)] \end{aligned}$$

We want to rewrite this causal quantity using things observable in the data.

Adjustment formula: Condition on X

Law of iterated expectation gives:

$$E[Y(1)] = E_x\{E[Y(1) | X]\}$$

$$E[Y(0)] = E_x\{E[Y(0) | X]\}$$

$$ATE = E_x\{E[Y(1)|X] - E[Y(0)|X]\}$$

Adjustment formula: Use exchangeability

$$(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$$

$$E[Y(1)|X] = E[Y(1)|T = 1, X]$$

$$E[Y(0)|X] = E[Y(0)|T = 0, X]$$

Within X , treated and untreated units represent the same potential-outcome distribution.

Adjustment formula: Use Positivity

To write $E[Y(1)|T = 1, X]$, we need:

$$P(T = 1|X = x).$$

Similarly, we also need $P(T = 0|X = x) > 0$.

Otherwise, for some x , $E[Y(1)|T = 1, X = x]$ has no data behind it.

Exchangeability tells us whom we may compare; positivity ensures both groups actually exist to make that comparison.

Adjustment formula: Use Consistency

If $T = t$, then $Y = Y(t)$

$$E[Y(1)|T = 1, X] = E[Y|T = 1, X]$$

$$E[Y(0)|T = 0, X] = E[Y|T = 0, X]$$

$$ATE = E_x[E(Y|T = 1, X) - E(Y|T = 0, X)]$$

From counterfactuals to observed data

$$E[Y(1) - Y(0)] - \text{Causal Estimand}$$



Exchangeability, Positivity,
Consistency, SUTVA

$$E_X[E(Y | T = 1, X)] - E[Y | T = 0, X]$$

This is identification: the causal estimand is now written using quantities available in the observed data.

This is standardization / the g-formula in its simplest form.

From counterfactuals to observed data

Identification

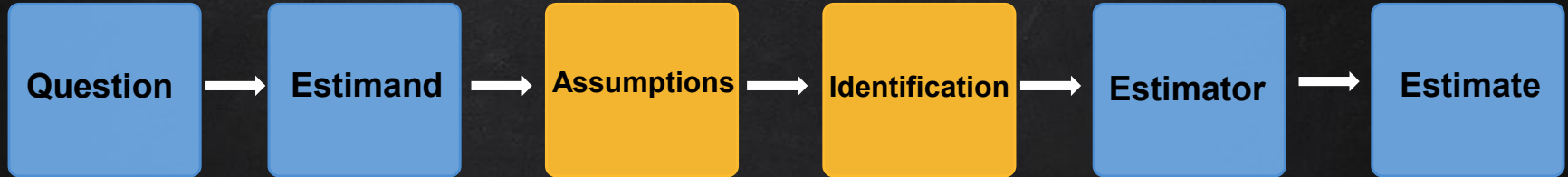
Can we express the causal estimand as a function of the observed-data distribution?

Estimation

How do we estimate that function from a finite sample?

The assumptions justify the observed-data target. Regression, matching, IPW, and machine learning are ways to estimate that target. They do not create causality.

Causal Flowchart



The order matters: define the question and estimand, justify identification, then choose an estimator.

Simulation Study

Outcome: systolic blood pressure

Treatment truly lowers BP by 5 mmHg for everyone

Older patients are more likely to receive treatment

Age also raises blood pressure

Therefore, treatment assignment is confounded by age.

In a simulation we can store both $Y(0)$ and $Y(1)$ — something impossible in real data.

Simulation Study

```
set.seed(123)

n <- 2000
age <- rnorm(n, 50, 12)

# Older patients are more likely to be treated
p_treat <- plogis(-4 + 0.08 * age)
T <- rbinom(n, 1, p_treat)

# Potential outcomes: treatment lowers BP by 5
Y0 <- 120 + 0.6 * age + rnorm(n, 0, 8)
Y1 <- Y0 - 5

# Only one potential outcome becomes observed
Y <- ifelse(T == 1, Y1, Y0)
dat <- data.frame(age, T, Y, Y0, Y1)
```

Simulation Study

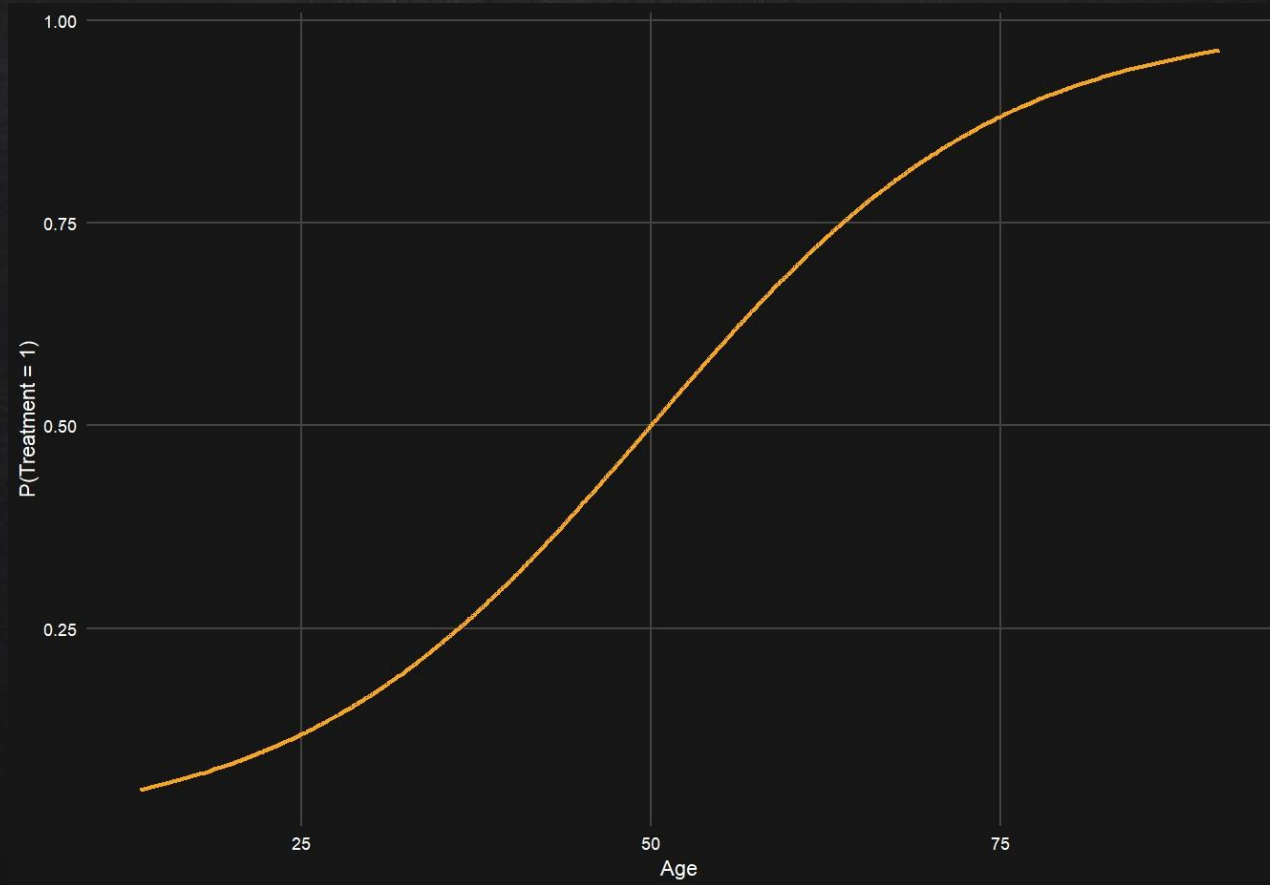
```
# True causal effect -- available only because  
this is simulation  
mean(dat$Y1 - dat$Y0)  
# [1] -5  
  
# What an observational analyst sees  
with(dat, mean(Y[T == 1]) - mean(Y[T == 0]))  
# about +1.34 mmHg
```

True effect: -5

**Naive difference:
+1.34**

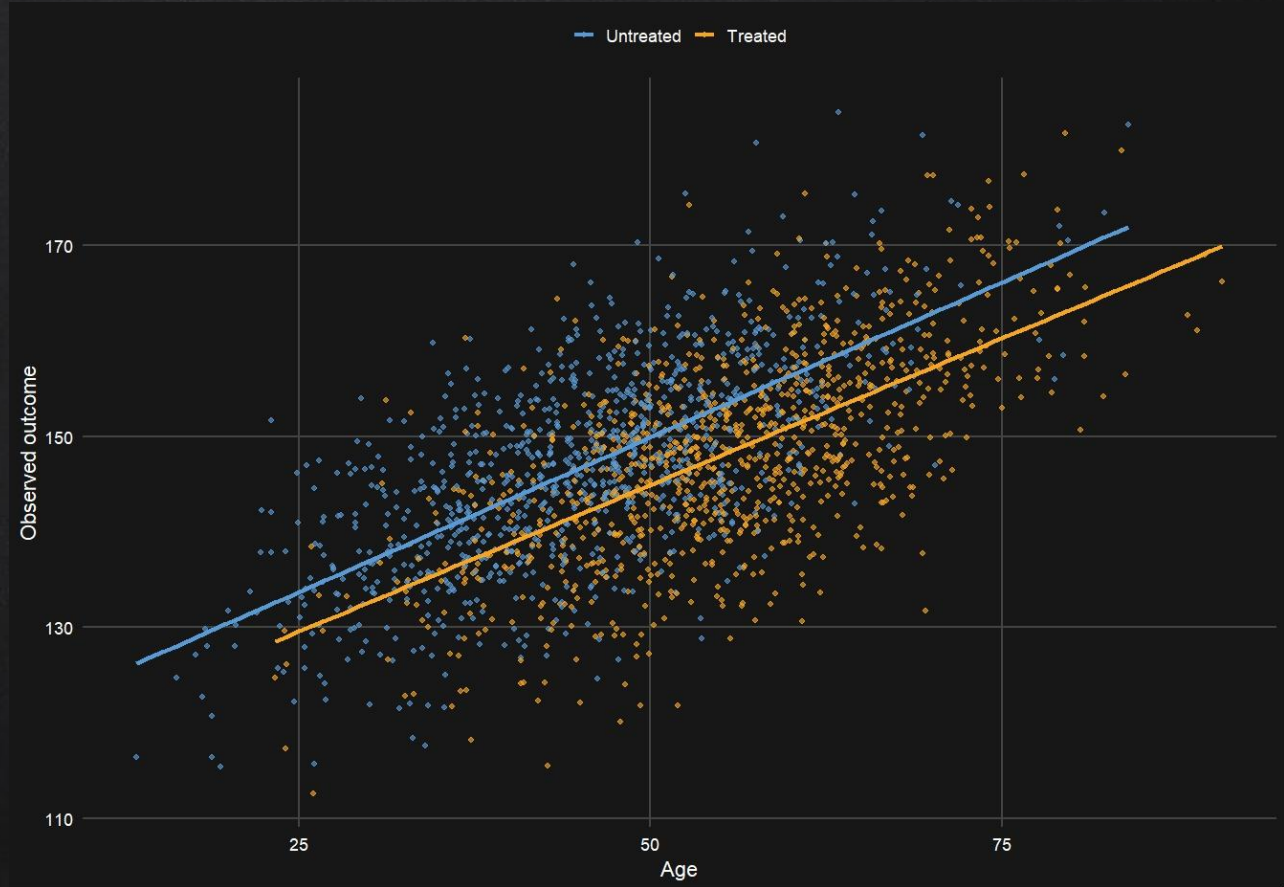
The sign is wrong.

Why naïve comparison wrong

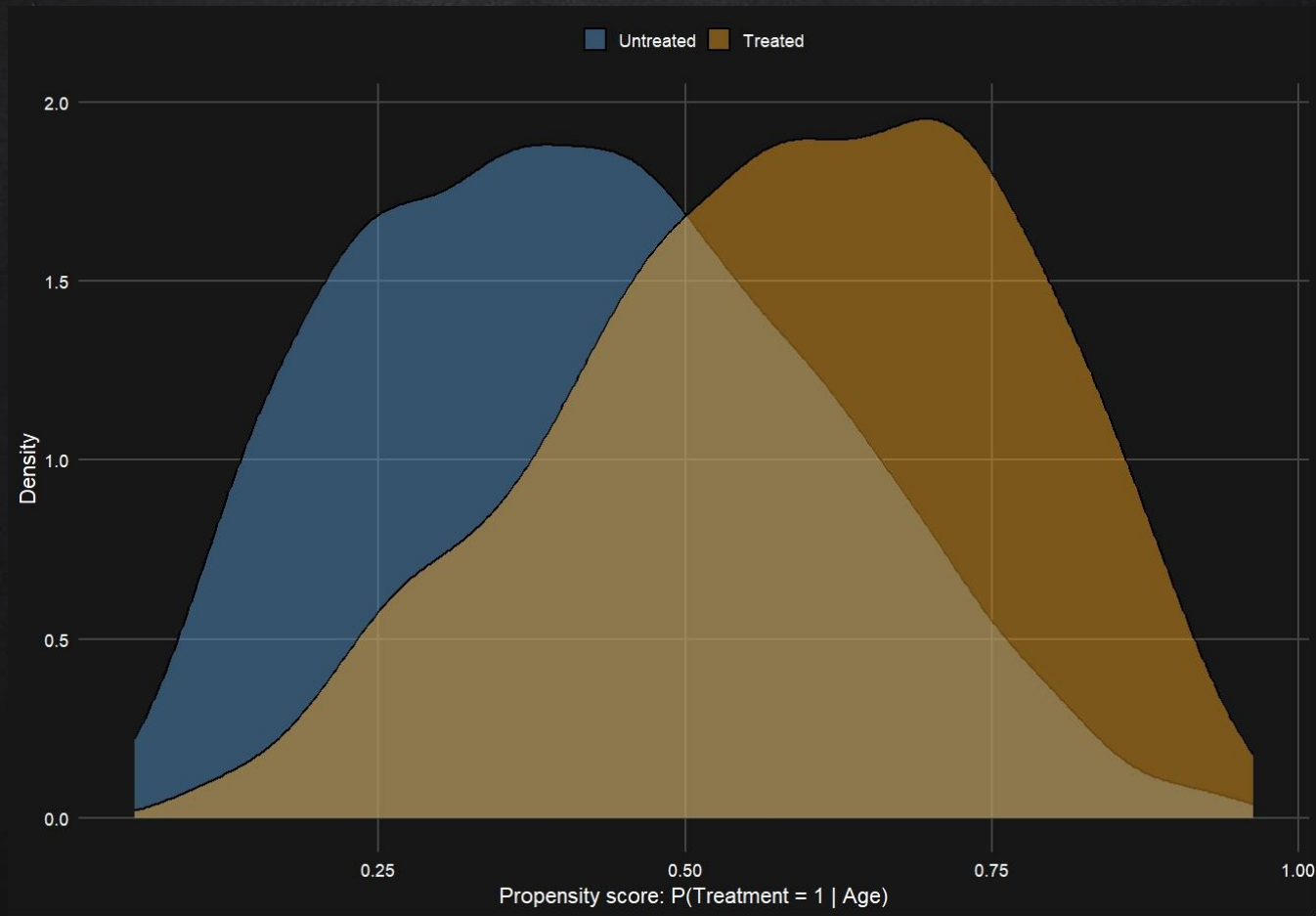


Treated and Untreated groups have different population

Older patients have higher BP and are more likely to receive treatment; the causal -5 mmHg effect is masked.



Overlap Check



Simulation Study

```
fit <- lm(Y ~ T + age, data = dat)
summary(fit)
```

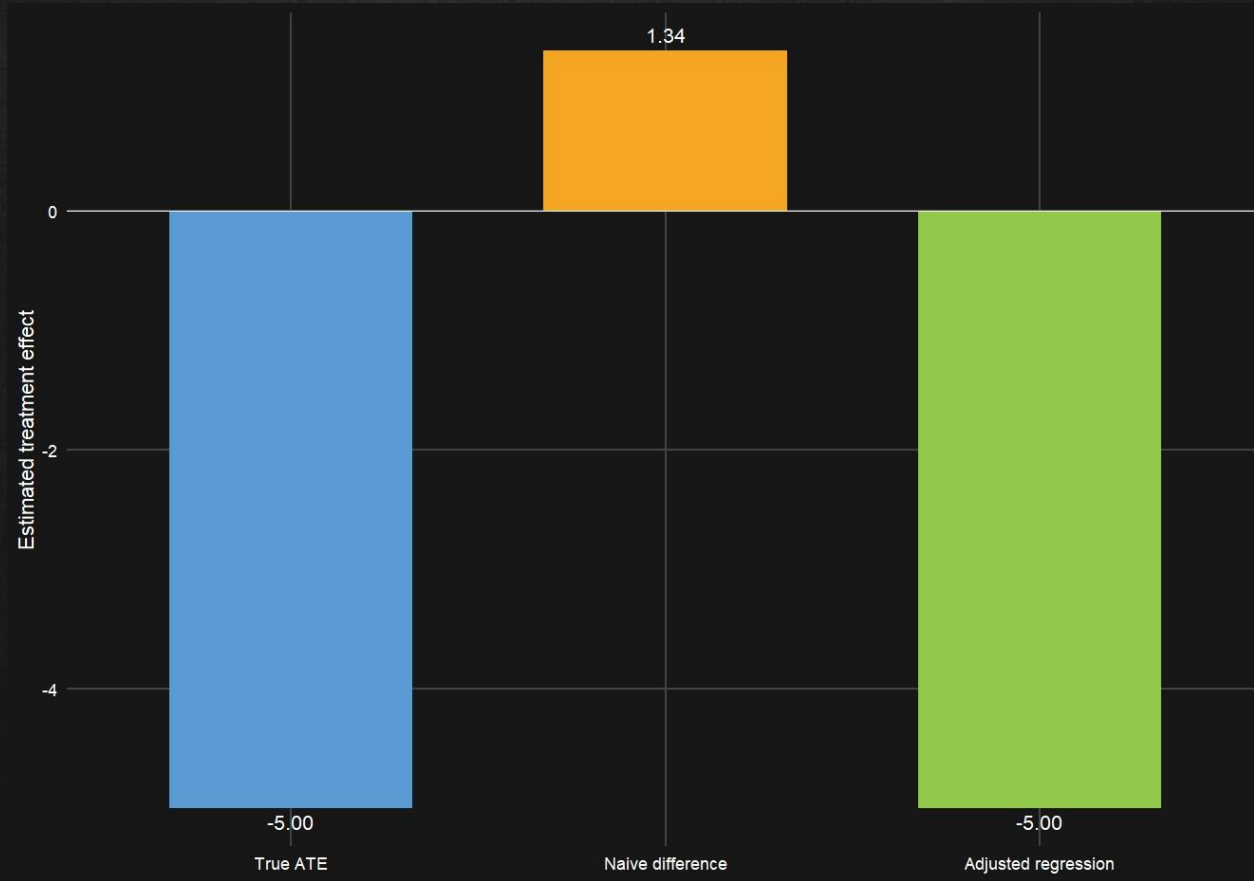
```
# Coefficient of T is approximately
#    - 4.998725
```

True effect: -5

**Adjusted estimate
 ~ -4.99**

Adjustment recovers the causal effect because the model and assumptions match the simulation.

Why naïve comparison wrong



Regression Coefficient is not automatically causal

***lm*($Y \sim T + X$) does not create exchangeability.**

Did we adjust for the right pre-treatment confounders?

Did we accidentally adjust for a mediator or collider?

Is there adequate overlap?

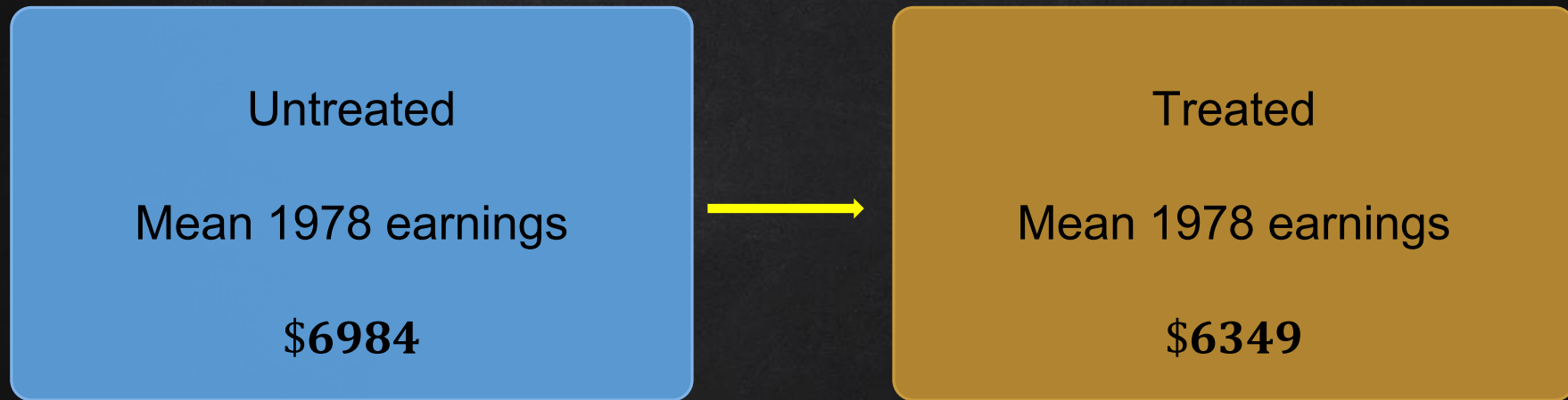
Is the functional form adequate?

Is the target estimand the regression coefficient we are interpreting?

Mediator: A variable on the causal pathway from treatment to outcome.

Collider: A variable caused by both treatment and another factor.

Real Data Analysis: LaLonde / NSW Job training



Question: Did job training increase later earnings?

Naïve Difference: $-\$635$

Real Data Analysis: LaLonde / NSW Job training

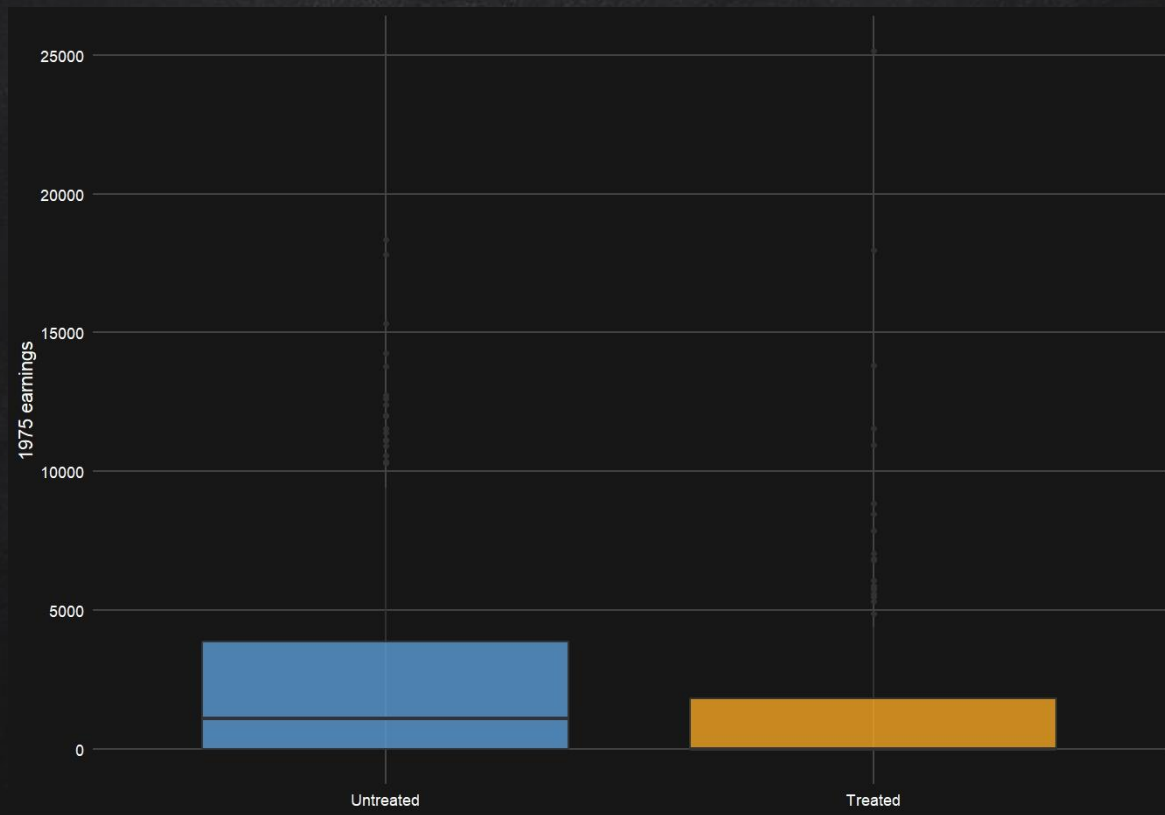
Baseline characteristics

Covariate	Untreated	Treated	What stands out?
Age	28.0	25.8	Treated younger
Education	10.2	10.3	Very similar
1974 earnings	\$5,619	\$2,096	-\$3,523
1975 earnings	\$2,466	\$1,532	-\$934
Married	51.3%	18.9%	Much lower
No degree	59.7%	70.8%	More common

Treatment participation was associated with characteristics that also predict later earnings.

So, the raw \$6,349 vs \$6,984 comparison mixes treatment effect with baseline differences.

Real Data Analysis: LaLonde / NSW Job training



Real Data Analysis: LaLonde / NSW Job training

Naive comparison

-\$635

treated minus untreated

Adjust for age, education,
race,
marriage, degree status, re74,
re75

Adjusted estimate

+\$1,548

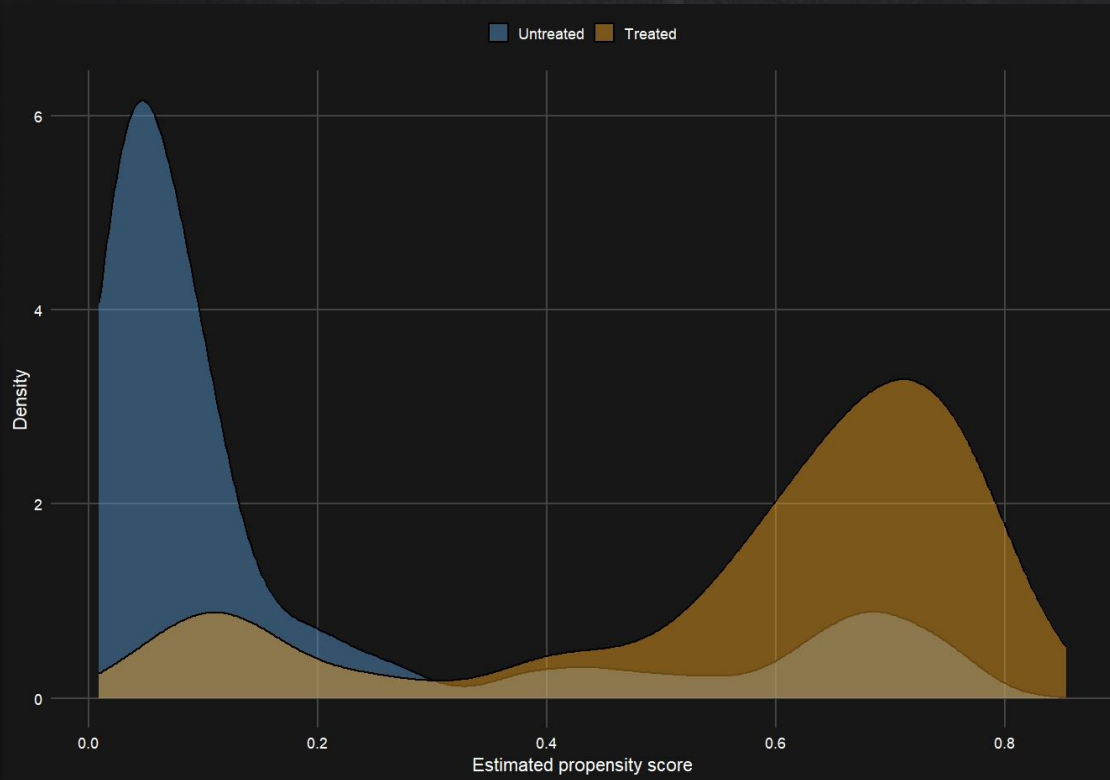
regression / g-computation

The sign changed from negative to positive after accounting for baseline differences.

That is exactly why “association” and “causal effect” are different questions.

Real Data Analysis: Overlap Plot

Propensity Score Distribution



What do we see?

Untreated participants are concentrated at low propensity scores.

Treated participants are concentrated at much higher scores.

There is some common support, but the overlap is limited.

Practical positivity concern

Real Data Analysis: LaLonde

What did we learn from the data analysis?

- The naive comparison suggested training decreased earnings by about \$635.
- Baseline characteristics showed strong differences between treated and untreated people.
- Regression adjustment changed the estimate to approximately \$1,548.
- But the propensity score plot revealed poor overlap.

Real Data Analysis: LaLonde

Therefore: \$635 is clearly a crude association.

\$1548 is an adjusted causal estimate under assumptions, but its credibility is limited by poor overlap.

Possible Solutions:

- Restrict to common support (Change the target population)
- Propensity-Score Matching
- IPW (Can be problematic with poor overlap)

If there is zero overlap, then no statistical adjustment method can recover the missing counterfactual from these data without additional assumptions.

Conclusion

Potential outcomes give us the language of causality.

$$Y(1), Y(0), ATE = Y(1) - Y(0)$$

But for each person, we only observe **one** potential outcome.

Moving from association to causation requires assumptions:

- **Exchangeability**: treated and untreated are comparable after conditioning on X .
- **Consistency**: observed treatment corresponds to the intervention we defined.
- **SUTVA / no interference**: one person's treatment does not alter another's outcome.
- **Positivity**: both treatment options are represented within relevant covariate profiles.

Identification comes before estimation. $E[Y(a)] = E_X\{E(Y|A = a, X)\}$.

A statistical model can always produce an estimate.

Causal inference asks whether the data and assumptions support interpreting the estimate causally.

Note: You cannot fully verify all causal assumptions from the observed data.